

# Where the Snap Fails

## *The Real Numbers as Counterexample*

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This companion document is written for general readers. It explains in plain language why the real number line cannot serve as the mathematical substrate for the Binary Snap, and why the 2-adic metric  $Q_2$  is required. No Lean verification is associated with this document; the relevant formal results are in ZP-B (p-adic topology) and ZP-C (information theory).

## I. The Natural Question

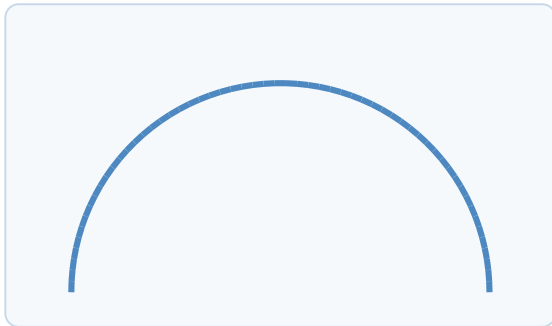
If you have read ZP-B, a natural question arises: why the 2-adic metric? Why p-adic numbers at all? The real numbers are more familiar and more widely used. They seem like the natural substrate for any framework involving continuity, limits, and state transitions. This document is the answer.

The answer begins with a counterexample. The real number line is not a valid substrate for the Binary Snap — not because it is wrong, but because it is the space where the snap is structurally impossible. Understanding why the snap fails in  $\mathbb{R}$  is the clearest path to understanding why  $Q_2$  is required.

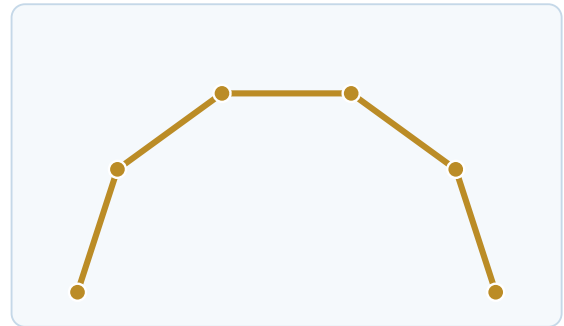
## II. The Density Symmetry

The smooth nature of the real numbers comes from the fact that numbers like  $\pi$  can have infinitely long decimal expansions. In that framework, there is no cleanly getting away from zero for the same reason there is no cleanly approaching it. The density is symmetric: approaching and departing from zero both have the same property. Between any two real numbers — no matter how close — there is always another one.

Zero in  $\mathbb{R}$  is not a special topological location. It is an ordinary point on a continuum that looks the same from every direction. For any candidate first step  $\epsilon_0 > 0$ , the number  $\epsilon_0 / 2$  also exists, and  $\epsilon_0 / 4$ , and  $\epsilon_0 / 2^n$  for every  $n$ . A discrete, irreversible departure from zero — the Binary Snap — is structurally impossible in  $\mathbb{R}$ .



Standard scale — smooth and continuous  
No first step away from zero



Zoomed in — the same arc shows discrete steps  
A minimum unit of departure becomes visible

The same mathematical arc — a piece of a circle, the curve that defines  $\pi$  — at two sampling resolutions. At standard scale it appears smooth and continuous (left). At high zoom, discrete steps become visible (right). In the actual real numbers, the decimal expansion can always continue; the smooth curve never structurally breaks down. The snap requires a space where that breakdown is built in, not imposed.

### III. Where the Infinity Lives

The distinction between  $\mathbb{R}$  and  $\mathbb{Q}_2$  is not a matter of degree. It is a question of where the infinity lives.

In  $\mathbb{R}$ , the infinity lives in the representation. An infinitely long decimal like  $\pi = 3.14159\dots$  describes a finite magnitude — a number sitting between 3 and 4. The infinite expansion is the notation, not the thing being noted. The number itself is finite and well-located on the line. Zero is a limit point: surrounded by non-zero reals, reachable as a limit but not isolated from them.

In  $\mathbb{Q}_2$ , the infinity is the address of zero. The 2-adic valuation assigns  $+\infty$  to 0 — not as a limit the valuation approaches, but as its actual value. Every non-zero element carries a finite integer valuation. The gap between infinite valuation and any finite valuation is not a limit. It is a structural discontinuity built into the metric itself.

| $\mathbb{R}$ Real Numbers                                                  | $\mathbb{Q}_2$ 2-adic Numbers                                     |
|----------------------------------------------------------------------------|-------------------------------------------------------------------|
| 0 is a limit point — surrounded by non-zero reals on all sides             | 0 is isolated — 2-adic valuation of 0 is $+\infty$                |
| Infinity lives in the representation (infinite decimal = finite magnitude) | Infinity is the address of 0 (the valuation itself is $+\infty$ ) |
| Departure from 0 is continuous — always subdivisible                       | Departure from 0 is a discrete jump in valuation                  |
| The snap cannot occur — density blocks every candidate first step          | The snap is a theorem — the valuation gap forces it               |

### IV. Finite Precision Forces the Snap

Every computer already knows this. Any number system with a maximum number of decimal places — any fixed-point arithmetic system, any discretized simulation — has a smallest representable positive number. Call it  $\delta$ . Below  $\delta$  you cannot go. The density argument fails at  $\delta$ . The snap is forced: there is a genuine first step, and halving it is not possible.

The real numbers are the idealization in which this floor is removed — the limiting case in which precision is unbounded and the minimum disappears. That is not a feature from the perspective of the Zero Paradox. It is exactly what blocks the snap.

$\mathbb{Q}_2$  is not an artificially truncated real line. It is a different metric space in which the floor is structural — not imposed by finite memory or finite precision, but built into the 2-adic valuation itself.

Every finite computational system is already subject to the Binary Snap by construction. The minimum representable positive value exists; the density argument fails there. The real numbers are the most familiar example of a structure that removes this floor — but any dense ordered set has the same property, including the rational numbers: for any  $\epsilon_0 > 0$ ,  $\epsilon_0 / 2$  also exists.  $\mathbb{Q}_2$  puts the floor back in, mathematically rather than by truncation.

## V. The Curve of a Perfect $\pi$

There is a physical version of this argument. At scales approaching the Planck length, current mathematical models including the real number line are understood to break down — what replaces them at that regime is an open question in physics, and the argument here does not depend on the answer.

## VI. Mathematical Constants and Forcing

A natural follow-on question: if infinitely long decimals are the issue, what about  $\pi$  itself?  $\pi$  is infinitely long. Does it already qualify for the incompressibility results elsewhere in the ZP framework?

No — and the reason matters.  $\pi$  is infinitely long in its decimal expansion, but it is computable. Finite algorithms — the Leibniz formula, the BBP algorithm, dozens of others — can generate  $\pi$  to any precision you specify, however large. There is no bound on the precision you can ask for. But this is the point, not a problem: the Kolmogorov complexity of the first  $n$  digits of  $\pi$  is tiny relative to  $n$ . The short algorithm is the information, not the infinite decimal. You can demand arbitrarily many digits; the specification of  $\pi$  remains short regardless.

This is what makes  $\pi$  a constant rather than an arbitrary number. It is the necessary consequence of a geometric relationship — the ratio of circumference to diameter — and that relationship provides the compression. The value is forced by the definition. The computability follows from the forcing. Constants are computable because they have finite definitions; the finite definition is the short program.

Most real numbers are not like this. Almost all real numbers — in the measure-theoretic sense — have no finite definition that picks them out. They are algorithmically random: no short program generates them; the first  $n$  digits have Kolmogorov complexity close to  $n$ . High algorithmic complexity is the signature of a number with no mathematical structure behind it, no definition that points to it specifically.

## VII. Two Kinds of Incompressibility

This brings us to a subtle but important point. ZP-C includes the result L-INF: the null state  $\perp$  has unbounded surprisal — no finite external description can capture it. A careful reader might ask: is this the same as saying  $\perp$  is algorithmically incompressible, like a random real number?

It is not. The distinction is the difference between randomness and necessity.

A random real number is incompressible because it has no structure — no mathematical relationship forces it to be what it is. Its complexity is high because it is arbitrary. Nothing points to it specifically.

$\perp$  is incompressible for the opposite reason: because anything standing external to it is structurally excluded. Nothing can occupy a position outside  $\perp$  to describe it — not because the description is too complex, but

because the position of "external to  $\perp$ " does not exist.  $\perp$  is not arbitrary. It is the most constrained object in the framework, the unique global minimum of the lattice. Its incompressibility is not the noise of randomness. It is the silence of structural necessity.

The real number line mixes both kinds — computable constants alongside algorithmically random reals — with no topological distinction between them. ZP-C's L-INF is a claim of an entirely different character.

#### The contrast in one sentence

A random real is incompressible because nothing forced it to be what it is.  $\perp$  is incompressible because nothing can stand outside it. One is the absence of structure. The other is structure all the way down.

## The Minimal Required Structure

The real numbers are the most natural, most familiar metric space in mathematics. They are also precisely the space where the Binary Snap cannot occur. Zero is a limit point, density is symmetric, and every candidate first step admits an infinite sequence of smaller steps below it.

$\mathbb{Q}_2$  is not an exotic choice. It is the minimal metric structure in which the structural isolation of zero is built in — not imposed by finite precision, not approximated, but present in the 2-adic valuation itself. The framework did not choose unusual mathematics for its own sake. It followed the result to the structure the result required.

#### Where the Snap Fails

The Binary Snap is impossible in  $\mathbb{R}$ : for any  $\epsilon_0 > 0$ ,  $\epsilon_0 / 2$  also exists, and the density argument shows no discrete departure from zero can occur.  $\mathbb{Q}_2$  is required because the 2-adic valuation structurally isolates zero — its valuation is  $+\infty$ , while every non-zero element carries a finite valuation. The gap between them is not a limit. It is the theorem.